AUDITING GENERALIZATION CLAIMS FOR LLM AGENT HARNESSES: SEMANTIC BINDING AND MEASUREMENT VALIDITY

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ABSTRACT

In our deliberately tool-green semantic-binding benchmarks, a retrospective audit of 169 frozen agent trajectories with hash-verified verdict-to-submission pairing found that family-specific tool-success signals accepted every run, including all 82 semantically incorrect bindings. Typed provenance/authority oracles separate these semantically invalid successes from real ones by verifying whether submitted values occupy the roles the evidence authorizes; across those trajectories the tool signal is constant, so all measured discrimination comes from the oracle. Measuring semantic binding this way, we ask how far a harness intervention transfers, and introduce a *claim-scope framework*. Generalization here is not one axis but three—over task instances, model backends, and task families—and widening one is *incomparable* to widening another, not a further rung: crossing families implies nothing about crossing models, and conversely. We therefore index the *evidence* for a claim by the set of (family, instance, model) configurations it occupies, ordered by inclusion, and require each dimension to be evidenced on its own. On a controlled *semantic-handoff* task, an instance-answer-independent "clarity bundle" (BundleS) is associated with zero observed axis-binding failures for Qwen3.7-Max where the ambiguous baseline fails two of three, and that direction reproduces on a pre-frozen held-out instance; a disclosure-only enumeration rules out the treatment simply publishing the answer. At three repetitions per condition each cell is imprecise (Fisher $p = 0.40$; exact interval 29.2–100.0% vs. 0.8–90.6%), so we report an *observed and once-replicated* effect, not an established one. Widening the model dimension (DeepSeek-V4-Pro under exact counterbalancing) and the family dimension (two independently executed prospective STA panels; a second family at a non-discriminative ceiling) establishes nothing further. Our primary family-widening evidence is the later, separately preregistered panel at k=6 repetitions per cell (216 episodes): -2.8 percentage points, moving over -31.9 to $+26.4$ under instance resampling. Raising repetitions resolved the instances rather than stabilising the aggregate—5 of twelve carry the whole contrast, with differences from -100.0 to $+100.0$ points in *both* directions, and 7 of 36 cells disagree across six identical repetitions—and that instance-level structure recurs in the earlier k=2 batch while the aggregate direction does not. What a harness claim has to report is therefore an instance-level response structure and a panel composition, not one transfer coefficient. The joint cell—different model *and* family—is measured with DeepSeek-V4-Pro at k=2 (72 episodes): $+12.5$ points, 5 instances improving to 2 declining, band -16.7 to $+41.7$, p=0.45. Joint transfer is *not established* either—not the same claim as no effect, on a panel where 4 of twelve instances cannot express a difference at all. A second, orthogonal axis—a four-layer *Harness Effect Audit Protocol* (capability, sampling, execution, artifact integrity)—caught a threat at every layer that would have changed the scientific conclusion. An external probe (Terminal-Bench 2.0→2.1, 26 repairs) finds the taxonomy relevant but incompletely covering (1 direct, 21 partial, 4 outside; sampling 0). Our instances test existence, recurrence, and transfer direction rather than estimate population pass rates.

1 INTRODUCTION

A commercial tool can sign off on a configuration whose values are bound to the wrong typed semantic roles: the tool is not malfunctioning—its answer is valid—but it answers a different question than the task poses, so tool success cannot decide task correctness. Our families are constructed so this is possible, and a retrospective audit shows real agents repeatedly enter that region (§3.1); typed provenance/authority oracles are what separate those cases, and are the grading substrate every number below already rested on. With a measurement that can see the failure, the question becomes how far an intervention that reduces it transfers.

When a harness intervention changes agent behavior, what can we claim? The question is usually answered along one implicit axis of "how general." But the unit a study actually realizes is a *configuration* $c = (f, i, m)$—a task family, a task instance belonging to it, and a model backend—so what indexes the *evidence* for a claim is the set of configurations at which it was measured, its **evidence support** $E \subseteq \mathcal{C}$, ordered by inclusion. The three familiar notions of generality are *projections* of that support:

- $s_I = \pi_I(E)$ **(instance scope):** the task instances a claim ranges over—$\{i_{\text{dev}}\} \subset \{i_{\text{dev}}, i_{\text{held}}\} \subset$ a declared frozen panel $\subset$ the generator's instance population.
- $s_M = \pi_M(E)$ **(model scope):** the backends—$\{$Qwen3.7-Max$\} \subset \{$Qwen3.7-Max, DeepSeek-V4-Pro$\} \subset$ a model population.
- $s_F = \pi_F(E)$ **(family scope):** the task families—$\{$workflow$\} \subset \{$workflow, STA$\}$, and so on up to a family population.

We index by the support and not by the triple (s_I, s_M, s_F), for two reasons. First, **instances are family-specific**: an STA instance is not a workflow instance evaluated elsewhere, so a family-widening probe necessarily brings its own new instance identities and s_I and s_F are not free coordinates of a product. Second, and consequently, $\pi_F(E) \times \pi_I(E) \times \pi_M(E) \neq E$ in general—a support is typically *sparse* in the product of its own projections—so having widened every projection somewhere is not the same as having measured every configuration. That gap is where this study's central omission lives, and a triple of marginals cannot express it. Finally, E *audits* a claim rather than stating it: a claim may name a wider **target support** T, and $E = T$ only when nothing is generalized. The lattice orders E; the standard of §3.1 governs which relations between E and T a design licenses.

Enlarging the support requires evidence *inside* the enlargement, and inclusion is a **partial order, not a chain**. The two ways a harness result is usually generalized—"it also works on another model" and "it also works on another task family"—add **incomparable** configurations: neither support contains the other, so neither is a prerequisite for the other, and an intervention can cross every family for one model while failing on a second model. Arranging them as successive rungs of a ladder would make a claim look better-indexed than its evidence, and would license the invalid inference that a cross-family result presupposes a cross-model one. Failure at a larger support does *not* refute a claim at a smaller one; it bounds the claim's scope.

Our frozen evidence realizes four of the five named scope regions below, and does so across five distinct supports (Figure 1): **S0** (local—Qwen3.7-Max, workflow family, development instance); **S1** (=S0 plus a pre-frozen held-out instance, widening s_I); **S2-M** (=S0 plus DeepSeek-V4-Pro, widening s_M); **S2-F**, which is a *class* of two incomparable family-widening supports (STA and SPICE) rather than one; and **S3**, any support holding a configuration whose family is not the workflow family *and* whose model is not Qwen3.7-Max. S2-M and S2-F are siblings, not consecutive levels. S3 is occupied by one measured support—DeepSeek-V4-Pro on the STA family—and, crucially, occupying it required running it: widening s_M and s_F separately does not reconstitute the joint support, so no combination of the other cells would have filled this one.

Because each S0/S1 cell rests on one instance at k=3, we call that result *observed and once-replicated* and reserve "established" for the standard of §3.1; Table 1 gives the numbers. Our primary S2-F evidence is a separately preregistered twelve-instance panel at k=6; an earlier k=2 panel over the same instances is reported beside it as an independent batch, never pooled with it.

A harness-effect claim has two independent questions: (A) *what scope of generalization does the evidence support?* and (B) *was the measurement itself valid?* The scope lattice addresses (A); a four-layer audit protocol (capability, sampling, execution, artifact integrity) addresses (B). Figure 1 places our concrete evidence and incidents into this 2D framework.

Contributions. (1) **Semantically attested evaluation.** Typed provenance/authority oracles, instantiated across three executable EDA task families, separate professional-tool success from correct semantic role binding; a retrospective, hash-paired audit of frozen trajectories shows the family-specific tool-success signal was *constant* over all 169 retained runs and accepted all 82 observed semantic misbindings (§3.1). Because the tasks are constructed so wrong bindings can remain tool-green, this quantifies real-agent occupancy of that region, not its prevalence in benchmarks generally. (2) The semantic-binding failure itself, in two subtypes we never collapse—axis-binding, and role-conditioned value selection—exhibited as real, tool-green agent trajectories. (3) A claim-scope framework indexing a harness-effect claim's *evidence* by the configurations it occupies rather than by a single ladder, making instance, model and family generalization distinct and partially ordered, making explicit what a study leaves unmeasured, and applying one qualification standard symmetrically to our positive and negative results. (4) A measurement-validity framework in which a verdict must be bound to the artifact it claims to judge and an oracle inherits the custody of its reference standard: capability, sampling, execution and artifact failures each materially changed scientific interpretation, plus external taxonomy evidence from Terminal-Bench repairs.

2 Related Work and Problem Formulation

Harnesses and harness effects. A *harness* is the task-level information structure visible to the agent (prompt, visible files, disclosure bundle, public tool feedback, action surface). A *harness effect* is a change in semantic-binding correctness attributable to harness information content, holding the hidden truth, grader, and tool fixed.

Problem formulation. Each task is a *semantic handoff*: bind a tuple to canonical typed roles from role-misleading evidence. The workflow family binds PVT descriptors to typed axes; Family A (STA/PrimeTime) binds (intent_class, target_partition, check_mode) from an authority provenance DAG; Family B (SPICE/HSPICE) binds (corner, load_condition, metric) from a request–authority join.

TAB vs. semantic binding. Task alignment (Mavali et al., 2026) asks *which* environmental evidence an agent should follow; semantic binding asks *which semantic role* selected evidence is permitted to fill. Our failures can occur after relevant evidence has been retrieved and can remain tool-green—the agent finds the right artifact but binds its value to the wrong typed axis.

Positioning. Yao et al. (2026) broadly characterize model×harness interactions; Wang et al. (2026a) critique harness-leaderboard conflation; Ursekar et al. (2026) optimize harnesses. We do *not* claim priority over the general observation that a reported success can be unearned; four recent works bound our claim, and what remains ours is narrower. Zhu et al. (2025) and Shao et al. (2026) set benchmark-construction and protocol-validity standards, the latter formalizing whether the target capability stays necessary for success and quantifying shortcut-driven score inflation—but our agents do not bypass the protocol: they complete it, a real commercial tool accepts the artifact, and the values still occupy the wrong roles. Gao & Zhou (2026) asks whether stored evidence supports a claimed outcome at all, labelling unsupported runs Unknown and widening score bounds accordingly; we address the complementary case in which the evidence is complete and the tool genuinely accepts the final artifact. Fan et al. (2026) is closest conceptually—semantic roles for tool arguments, provenance tracking, role-specific authority contracts—but targets runtime security enforcement rather than benchmark measurement, so semantic roles plus provenance is not ours to claim as new. Luo & Peng (2026) audits whether a correct outcome drew on permitted information; our object is value-to-role authority binding, not target-answer exposure. Ma et al. (2026) report cross-benchmark harness gains, which we use as evidence that some interventions do transfer—so transfer must be demonstrated rather than assumed. Three further concurrent works and the transfer-dimension literature are positioned in Appendix K; against all of them, what we add is the claim-scope lattice, a controlled semantic-binding mechanism, and an audit of which measurement failures changed the scientific conclusion.

Classical measurement theory. The claim-scope lattice is not a new theory of generalization; it operationalizes an old one for agent harnesses. Generalizability theory already indexes a measurement's reach by explicitly declared facets, each named fixed or random relative to a stated universe of generalization (Brennan, 2001); external validity already asks whether a causal conclusion survives changes of persons, settings, treatments and outcomes (Shadish et al., 2002); and validity in the Messick tradition is a property of the inferences drawn from scores rather than a label a test owns (Messick, 1995), a reading recently brought to LLM benchmarks directly (Bean et al., 2025). What we add is the instantiation—model, instance, family and scaffold as operationally distinct widening steps—plus one structural difference that forbids the classical estimator. Over models and instances a universe is arguable; over *families* there is none, since we authored all three, so a variance component for "family" would be a number without a population. That is why we report each support separately and never pool them into a generalizability coefficient.

3 Methodology: The 2D Framework

The claim-scope lattice (first axis). The support E and its partial order are defined in §1. Two consequences govern the studies below: a study may legitimately measure S2-F without S2-M, but may present neither as subsuming the other nor treat either as evidence about the joint support S3; and because instances are family-specific, a cross-family verdict is never a pure single-projection move—a property of the design space rather than a confound, which we flag where it matters (§3.1).

The measurement-validity axis (second axis). Any measurement has four distinct validity dimensions: (1) capability validity (does the task measure the intended capability, including tool-green semantic alternatives?); (2) sampling validity (repeated trajectories, counterbalancing, membership arbitration); (3) execution validity (transport, terminal-vs-recovered, tool health, protocol completion); (4) artifact integrity (action-surface isolation, canonical-source integrity, custody). These are complementary audit dimensions, *not* independent factors and *not* an ordering: they interact, and a fault in one routinely presents as a symptom of another—our canonical-source mutation (Layer 4) surfaced as an apparent tool-health failure (Layer 3), which is exactly why the protocol records which layer a fault actually belonged to rather than which layer raised the alarm.

Figure 1: The 2D framework: claim scope (columns) $\times$ measurement validity (rows). Columns are supports in the claim lattice, *not* a progression: S1, S2-M and S2-F each enlarge S0 along a different projection, so S2-M and S2-F are incomparable siblings. S3 enlarges both s_M and s_F; widening each separately would not have filled it, so it was run. Its verdict is *not established*, for S2-F's reason—a finite panel with bidirectional instance-level structure—at a shallower repetition depth.

	S0 local	**S1** $+s_I$ held-out	**S2-M** $+s_M$	**S2-F** $+s_F$	**S3** $+s_M+s_F$
Cap.	tool-green wrong binding; typed oracle	BundleS 3/3 (direction replicated)	DeepSeek 3/4=3/4	STA n=12 at k=2 and k=6: not established; SPICE ceiling	DeepSeek on STA, n=12 at k=2: $+12.5$ pp, not established
Samp.	position-balanced blocking; non-det.	pre-frozen held-out inst.	counterbalanced block	two prospective 12-inst. schedules	*not measured*
Exec.	SSE streaming (censoring caught)	recovered degradation kept	—	SPICE action-surface repair	*not measured*
Art.	canonical-tree guard	custody hashes	—	—	*not measured*

3.1 Claim qualification: what "established" is allowed to mean

Indexing a claim by scope still leaves open how strong the evidence at a given support must be before a word like "established" is used. We state the standard in advance and—this is the part a claim-auditing paper most owes its reader—apply it symmetrically, at the same bar for our positive result as for our negative ones. Four tiers, each stated at length in Appendix E: **observed** (a descriptive separation at that support, no target beyond E); **replicated** (the direction reappears at *newly frozen* configurations enlarging the projection claimed—re-running the same configurations later is repeated measurement, never replication); **estimated on a fixed panel** ($T = E$, accompanied by a sensitivity band rather than a confidence statement); and **generalized** (the one tier asserting $T \supsetneq E$, requiring a distribution the design instantiates).

Under this standard the BundleS result is *observed* at S0 and *replicated* at S1—a replication in s_I, since S1 adds a new instance rather than re-running the old one; it is *estimated* nowhere in the workflow family, since each cell is one instance at k=3 (intervals 29.2–100.0% against 0.8–90.6%, two-sided Fisher $p = 0.40$), and it is *generalized* nowhere at all. The S2-F panels are *estimated* and not generalized. Table 10 (Appendix E) maps evidence shapes to the wordings they license; every verdict below sits inside it.

Construct: what semantic correctness means. Semantic correctness is *not* inferred from tool success, a hidden numeric answer, or the artifact score. It is decided solely by whether the submitted tuple matches the binding attested by the independent provenance/authority oracle. *Worked example (Family A):* visible evidence attests intent=*functional_close*, partition=*core*, check_mode=*setup*; the agent submits (functional_close, core, *both*)—PrimeTime returns green, the value is type-valid, but the coverage authority attests *setup*, not *both*; the oracle rejects it as a role-conditioned value-selection failure. This construct is instantiated independently in the workflow family (PVT axes), Family A (intent/partition/check_mode), and Family B (corner/load/metric), each with its own grader. We do *not* claim human validation (Study B was preregistered but unexecuted).

A worked instance (workflow family, ambiguous variant). The agent receives an out-of-date timing sign-off handoff and must repair `flow_config.json` to the unique package (netlist, clock, scenario, corner), then regenerate a two-stage evidence chain. The difficulty is deliberately *not* retrieval: the reports label their axis fields `op_point` and `mode`, and this variant ships no glossary or axis declaration, so which logical axis each label denotes must itself be recovered from the intersection of the sources. Five constraints are recoverable from the specification—K1 netlist family, K2 clock identity (the only clock with non-zero intended-clock coverage), K3 scenario-typed, K4 corner-typed, K5 the sign-off pair. (The frozen task files number these C1–C5; we relabel them K1–K5 because the clarity components of §4 are also C-numbered, and K5 is what C6 asserts.) Exhaustive enumeration over the declared domains yields 294 candidate assignments, of which **exactly one** satisfies K1–K5: (`netlist_v2.v`, `clk_main`, `slow`, `func`).

Four evidence artifacts are shipped, each *pairwise plausible* and each violating a different constraint (Table 3, Appendix A). Source A is the critical one: it carries the right netlist *and* the right clock, and **PrimeTime signs it off green**—but its two role fields are swapped, so a corner value occupies the scenario role and a scenario value the corner role. Every shipped source signs off green, so no tool signal separates A from the truth—only the typed oracle does. This is what *tool-green* means throughout the paper, and it is why a benchmark that scores this task by tool exit status would score a wrong binding as a pass.

How the oracle adjudicates. The grader never consults tool exit status when deciding semantic correctness. It applies typed-membership predicates to the submitted package: the scenario field must belong to the scenario axis and the corner field to the corner axis, neither may be a PVT descriptor (a descriptor denotes a scenario–corner *pair* and is never a value of either axis), and the clock must match by exact identity, so a generic alias is rejected. These predicates are folded into the master evidence gate, so a sign-off-green but mis-typed package cannot pass. A failure is classified into exactly one of two subtypes we never collapse: an **axis-binding failure**, where a value occupies the wrong typed axis (e.g. `func`/`slow`), and a **role-conditioned value-selection failure**, where the value is axis- and type-valid but the wrong member fills the role (e.g. `typ`/`func`). Uniqueness is verified by exhaustive enumeration before an instance is admitted, so "correct" is fixed by the instance's constraint system, not by grader judgement.

What the oracle discriminates that the tool does not. The oracle's necessity is usually argued from construction—a wrong binding *can* stay tool-green. A **retrospective, post-freeze audit** (specified after the program closed, not preregistered, adding no episode, model call or tool run) measures the consequence. Pairing each episode's semantic verdict with its family's own pre-existing tool-success field, and excluding rather than imputing 33 of 202 considered trajectories, the tool signal equals *accept* for all 169 retained runs across five strata—so all 82 semantic errors are accepted. Against the target variable it is a constant with **zero empirical discrimination** between the 87 correct and 82 incorrect submissions, and all discrimination the study reports comes from the oracle. This is not a population false-accept rate: strata are never pooled, and the families are built so a wrong binding survives the tool. Nor is it two findings—with the signal constant, $\Delta \equiv 1 - S_{\text{semantic}}$. What construction alone does *not* give is behavioral: a region reachable in principle was occupied by real agents 82 times. SPICE phase-5D is the negative control (18/18 correct, oracle and tool agreeing). Appendix G gives the per-stratum table, exclusions, and pairing sensitivity.

BundleS (canonical labels, value-domain definitions, glossary, procedural contract; component C6 withheld; no golden values); **TypedContract** (same information as BundleS as a JSON Schema). Hidden truth and grader are identical across conditions. We call BundleS **instance-answer-independent** rather than "non-answer-bearing": it withholds the instance's target tuple but deliberately discloses *task-level role semantics*—which axis a label denotes and what values each axis admits—so calling it answer-free would understate it. The same schema serves every instance, so it contains no explicitly instance-specific assignment field; whether schema and generator structure *jointly* permit recovering an assignment is a separate question, settled by the disclosure-only enumeration below.

Design and models. Seeded exact-counterbalanced blocked randomization, frozen before any paid call. Instance is the primary unit; repetitions nested. We evaluate Qwen3.7-Max and DeepSeek-V4-Pro; these are not representative of the frontier-model population. The design is diagnostic, not population-estimating: it tests existence, recurrence, ceiling/null behavior, and transfer direction; it cannot resolve small effect sizes.

4 Study I — Local Effect and Held-Out Replication (RQ1; S0–S1)

On the workflow controlled pair (identical truth and grader; ambiguous vs. clear handoff), BundleS was associated with *zero observed* axis-binding failures for Qwen3.7-Max where the ambiguous baseline failed two of three: Base 1/3 $\rightarrow$ BundleS 3/3. On a *pre-frozen held-out* instance the direction reproduces: Base 1/3 $\rightarrow$ BundleS 3/3, again with no axis-binding failure under BundleS. Each is one instance at k=3 and therefore individually imprecise—exact intervals 0.8–90.6% for Base against 29.2–100.0% for BundleS, two-sided Fisher $p = 0.40$ at S0 and $p = 0.40$ at S1—so the evidential weight lies not in either cell's separation but in the direction reappearing at an independently frozen instance. The subset producing it withholds C6, and a **disclosure-only enumeration** settles the alternative explanation that withholding is meant to secure (Appendix F). Reading only what a condition discloses and never the task evidence, between 9 and 147 of the 294 candidate assignments stay compatible with BundleS—the interval spans whether its typed-axis membership counts as stated value content—so under *neither* reading does BundleS identify the assignment by itself, at S0 or at the pre-frozen held-out instance. The deliberately answer-bearing C6, by contrast, collapses the sign-off role pair to one under both readings. The effect is therefore not attributable to direct answer disclosure. It is not an equal-information comparison either: BundleS reduces semantic ambiguity by design, on the leakage-favourable reading narrowing the typed grid from 49 candidates to 9, so what the probe bounds is the treatment's informational advantage, not the behavioral mechanism producing the result. The failure taxonomy partitions the workflow ledger (70 episodes) into 41 correct and two subtypes we never collapse—24 *axis-binding* and 5 *role-conditioned value-selection*—all tool-green, rejected only by the typed oracle.

A sequential decomposition tested whether a stable minimal component underlies BundleS. None was isolated: C1 alone, C2-only, C4-only, and a C24 bridge all failed their thresholds. We therefore describe this as an *exploratory component localization that did not converge*, not as a demonstration that no component mechanism exists—failure to identify is not absence, and at k=4 per condition the design has little power to distinguish the two. Table 4 (Appendix B) gives the full per-condition ledger behind both statements.

Three things in that ledger are worth reading directly. First, the *full* bundle—which does include C6, the component that discloses the instance's assignment—moves both models from failure to 3/3; the scope limitation we report at S2-M is about the instance-answer-independent subset, not about the phenomenon. Second, the axis column is where the mechanism lives, but it does not localize to one ingredient: every condition that suppresses axis errors carries value-domain information, yet C2-only and C4-only carry it too and still retain three axis failures each, and the C1-alone row refutes the pre-declared hypothesis that canonical labels alone eliminate axis errors. Third, three condition–model pairs were measured in more than one run window and every one disagrees with itself: C24 gives 3/4 with no axis error then 2/4 with one, Base(0009)/DeepSeek gives 0/3, 2/3 and 3/4, and BundleS/DeepSeek moves from 1/3 to 3/4. That within-condition spread *across* windows, visible only because the ledger does not deduplicate repeats, is direct evidence that a component result at k=4 is not stable enough to name a mechanism. One caveat: those windows span 22 days and our records pin the *requested* alias, so stochasticity and backend drift cannot be separated (§6).

Verdict at S0 and S1: *observed*, and *replicated* once (§3.1); not *established*, and nowhere *generalized*. The separation is descriptive within each cell, and the replication rests on one independently frozen instance, so neither licenses a family-level treatment-effect claim; we make no universal improvement claim. A negative result at S2-M or S2-F does not refute this—it bounds the coordinate along which the effect has been shown, which is the entire purpose of indexing the claim.

5 STUDY II — WIDENING THE MODEL AND FAMILY COORDINATES (RQ2; S2-M, S2-F)

Does the S0–S1 effect survive widening s_M or s_F? Table 1 is the main result. The two questions are asked separately because they are separate coordinates: a verdict at S2-M constrains nothing at S2-F.

Table 1: Main result matrix (semantic-binding correctness; instance is the unit). Workflow/SPICE entries are correct of k stochastic repetitions; STA entries are the mean correct-rate over 12 instances. Verdicts use the qualification standard of §3.1 and Table 10: "observed" and "replicated" are deliberately weaker than "established." **S2-F is a class of supports, not one**—STA and SPICE have disjoint instance sets and neither family projection contains the other. **The three STA rows are independently executed batches over the same twelve frozen instances and are never pooled**: nothing is summed, averaged or differenced across them, and no row is evidence about another's execution conditions.

Family / model	Base	BundleS	TypedContract	#inst.	verdict
Workflow / Qwen3.7-Max (S0)	1/3	3/3	—	1	observed
Workflow / Qwen3.7-Max (S1, held-out)	1/3	3/3	—	1	replicated once
Workflow / DeepSeek-V4-Pro (S2-M)	3/4	3/4	—	1	not established
STA / Qwen3.7-Max (S2-F, k=6)	**0.278**	**0.250**	**0.361**	**12**	**not established**
STA / Qwen3.7-Max (S2-F, earlier k=2 batch)	0.208	0.333	0.458	12	not established
SPICE / Qwen3.7-Max (S2-F)	1.00	1.00	1.00	3	ceiling; uninformative
joint model×family (S3)	—	—	—	72	not established

S2-M (widening the model coordinate). Under exact counterbalancing, BundleS is not established for DeepSeek-V4-Pro (3/4 = 3/4; exact interval 19.4–99.4% in both arms, two-sided Fisher $p = 1.00$). This does not refute the S0–S1 result for Qwen3.7-Max; it bounds the model coordinate. Neither does it establish absence: at k=4 that interval is wide enough to contain effects of practical size, so "not established" here means the design could not resolve the question, not that the answer is no.

S2-F (widening the family coordinate, STA). Two prospective panels were run over the same twelve frozen instances: an earlier batch at k=2 and—because two repetitions cannot resolve a cell's own value—a later, separately preregistered batch at k=6 (216 episodes). **The k=6 batch is our primary S2-F evidence;** the k=2 batch is retained as the earlier study that motivated raising k, and the two are never pooled (Table 1). At k=6 the realized panel difference is -2.8 percentage points (BundleS $-$ Base), with 2 of 5 non-zero instance differences positive; the preregistered exact sign test gives $p = 1.00$, its permutation sensitivity $p = 0.72$, and resampling the twelve instances moves the estimate over -31.9 to $+26.4$. That band is a panel-composition sensitivity and not a confidence interval—the declared estimand is the realized panel, whose composition has no sampling distribution (Appendix C states the limits of the reading). *No consistent BundleS advantage is established at either repetition level*, and as at k=2 a p of 1.00 must not be read as evidence *for* a zero effect: it is a non-establishment, and the frozen analysis remains the sole basis for that verdict.

What raising k bought was not a tighter aggregate but a resolved one. Only 5 of the twelve instances can express a condition difference at all (6 at the floor in both arms, 1 at the ceiling), and across those five the difference runs -100.0 to $+100.0$ points—two instances at the largest magnitude the scale allows, in *opposite* directions. Because the per-instance effects sharpened rather than shrank, the composition band came out *wider* than the k=2 batch's: repetitions bound what a cell is worth, not which instances a panel contains. And 7 of 36 cells have six identical repetitions that disagree among themselves, so a cell estimated from one trajectory estimates a value the cell does not have. That structure is itself stable across batches (post hoc, not preregistered; Appendix C): 10 of 12 instances fall in the same class and all 4 informative in both agree in *sign*, so the bidirectional heterogeneity is not wholly an artifact of low repetition. That comparison pools nothing, and since both batches re-run the same frozen instances it is repeated measurement of stability, never replication in the sense of §3.1. TypedContract is again descriptively highest (0.361) with no advantage established—hypothesis-generating only. *Panel size matters separately from repetition:* the three-instance pilot (-16.7pp) and the twelve-instance k=2 panel ($+12.5$pp) ran in opposite directions—very small panels being unreliable for directional harness claims, not evidence that BundleS transfers (Wang et al., 2026b).

S2-F (SPICE). Base = BundleS = TypedContract = 1.00—a non-discriminative ceiling. This is a measurement limitation, not evidence against transfer.

S3 (joint model$\times$family). The same twelve frozen instances and three conditions, DeepSeek-V4-Pro at k=2 (72 episodes; Appendix C): Base 0.250, BundleS 0.375, TypedContract 0.417; $+12.5$ points, 5 instances improving to 2 declining, k^{+}=5 of 7 at p=0.45, band -16.7 to $+41.7$. The point estimate favours BundleS and the discrimination reaches no conclusion, so joint transfer is *not established*—not "no effect", with 4 instances at the floor in both compared conditions and 1 at the ceiling. Two limits bind hardest here. At k=2 a cell is estimated twice on a family where 7 of 36 cells were measured to disagree across six repetitions, so this aggregate is not set beside the k=6 one as though equally resolved. And while the analysis was fixed before any of these outcomes was read, the arm's *execution* was not preregistered; we keep those apart rather than let the first imply the second (Appendix D). Post hoc, the two panels show a descriptive instance-level concordance: of the 5 instances informative in both, 4 share the BundleS$-$Base direction. Model and k change together and that subset was identified afterwards, so this does not establish model-invariant instance structure.

Four outcomes, kept distinct. Widening s_F, and then s_M jointly with it, yields four things we never pool: *no established effect on a discriminating family* (STA at k=2 and k=6); a *non-discriminative ceiling* (SPICE); *no established joint transfer* (S3 at k=2); and *instance-level heterogeneity large enough in both directions that the panel mean is a composition statistic*. Neither non-establishment rules out a small effect. Nothing is summed across the three STA batches: no n=24, no k=8, no pooled episode count.

6 Study III — Measurement Validity and External Scope Probe (RQ3)

Each headline conclusion could have been manufactured or masked by a measurement artifact. The four-layer protocol is instantiated and stress-tested; Table 2 lists incidents where a control *materially changed the interpretation*.

Table 2: Audit-protocol incidents: each control prevented or corrected a wrong scientific conclusion.

Threat	Control	Conclusion that would have been wrong
Tool-success proxy (Cap)	Typed provenance/authority oracle	Tool-green wrong bindings scored as correct
Non-streaming censoring (Exec)	SSE + terminal-transport arbiter	Qwen3.7-Max falsely recorded as failing
Position confounding (Samp)	Exact position-balanced blocking	Base/BundleS order leaked into contrast
Recovered degradation (Exec)	Terminal-valid vs recovered split	Hard-but-recovered solve discarded
PT corrupted measurement (Exec)	Tool-health sentinel + bookends	Tool hiccup read as model failure
SPICE action-surface false positive (Art)	Immutable core + forensic audit	12 SPICE episodes spuriously zeroed; semantic-binding ceiling obscured
Canonical-source mutation (Art)	Exact-commit worktree + hash guard	Golden rewrite read as tool outage

A verdict is only about the artifact it can be shown to describe. Comparing a tool verdict with a semantic one presupposes both describe the same submission, which the workflow family does not guarantee: the agent may edit its configuration after running the evidence chain. Hash attestation excluded 9 of 58 workflow runs where tuple equality caught two, so a *post hoc* paired analysis must re-establish that binding (Appendix G; STA and SPICE satisfy it by construction).

A monitor is only as trustworthy as the custody of its reference standard. A monitor inherits its reference artifact's custody: if the reference is mutable by the harness, it faithfully reports a genuine mismatch while misattributing the cause—as ours did for a day, reading our own writes into the canonical tree as a remote outage (Appendix J). Hence a fourth Layer-4 requirement: *the reference standards of health checks must be isolated from the harness whose health they certify.*

Model custody: a fifth Layer-4 requirement, stated as our own gap. A measurement is reproducible only if the *system under test* is identified as precisely as its artifacts, yet a commercial API names it by a provider alias. A provider-resolved snapshot, response id or fingerprint survives for 0 of the 70 ledger episodes—the transport record has no such field, so the gap is total—and only 66 retain even the *requested* alias. A repointed alias means two measurements of the "same" condition need not measure the same system, so we add that requirement—*record the provider-resolved model identity per episode*—as our own gap: across the windows of §4 we cannot separate stochasticity from backend drift. Under it we disclose that the Phase-8A batches reached the same aliases through a *different serving endpoint*, the original having ceased to serve these models. We held the API parameters we control but cannot certify the serving implementation identical, so the change is disclosed and *not* treated as a factor: the batches are reported separately as independent executions, not because their endpoints differed.

External taxonomy scope probe (Terminal-Bench 2.0→2.1). Coding the 26 documented repairs under a rubric *frozen before coding* gives **1 direct, 21 partial, 4 outside** (capability 17, sampling 0, execution 10, artifact 1; 6 multi-layer): relevant but incompletely covering, not validation—static coding cannot test runtime portability and leaves sampling untested.

7 Limitations and Discussion

Controlled executable tasks in EDA, not sampled production incidents. Small instance counts—S0, S1 and S2-M each rest on one instance at k=3–4; no cell reaches the *generalized* tier of §3.1. *Construct validity*—the oracles are internally consistent and instance-unique but not shown to agree with expert judgement (Study B, preregistered, unexecuted). *Model scope*—two backends, named only by provider alias (§6). *Family scope*—three families, all ours and all within EDA, so S2-F reads as cross-generator transfer inside one designed universe. *Harness scope*—"harness" is the task-level information structure of §2, not the agent scaffold: every episode ran through our own runner, and since failure fingerprints are scaffold-specific though largely model-independent (Vats & Golev, 2026), the 82-of-169 occupancy is a property of that scaffold—a tool-success field's inability to discriminate role binding is not. *Audit protocol*—the layers were *derived from* our own incidents, so Table 2 is framework-construction evidence, the Terminal-Bench coding its first external test. *No mechanism isolated*—a localization that did not converge, not a demonstration that none exists.

What the higher-replication batch changes. It overturns nothing and establishes nothing; it changes which object is reportable. With 7 of 36 cells inconsistent across six identical repetitions, an aggregate built from one or two trajectories per cell is assembled from values its cells lack—and resolving them widened the band rather than narrowing it.

8 Conclusion

Tool success discriminated semantic correctness nowhere in our 169 retained trajectories, accepting all 82 misbindings; typed oracles carried the measurement. We therefore index a claim's evidence by its configuration support: instance, model and family generalization are partially ordered projections, not rungs—observed at S0, replicated once at S1, unestablished at S2-M, S2-F and, measured rather than inferred, S3. Across batches the aggregate moved and the instance-level structure largely did not (post hoc; nothing pooled), so what limits every widening claim is that structure and the panel's composition, not run noise alone. Each audit layer changed a conclusion, our own model custody weakest. Transfer must be demonstrated at each coordinate, not inherited (Ma et al., 2026).

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A WORKED INSTANCE — SHIPPED EVIDENCE SOURCES

Table 3: The four shipped evidence sources of the worked instance. Each is pairwise plausible and each signs off green under PrimeTime, yet each violates a different typed constraint: A and D swap the two role fields, B is out of the netlist family, C puts a PVT descriptor in the corner role and a generic alias in the clock role. The hidden role mapping (`op_point`→scenario, `mode`→corner) is not disclosed in this variant.

source	netlist	clock	op_point	mode	violates	tool
A role swap	`v2` ✓	`clk_main` ✓	`func`	`typ`	K3, K4	green
B stale	`v1` ✗	`clk_main` ✓	`slow`	`func`	K1	green
C PVT label	`v2` ✓	`clk` ✗	`slow`	`slow_1.0V_125C`	K2, K4	green
D JSON chain	`v2` ✓	`clk_main` ✓	`func`	`typ`	K3, K4	green
unique truth	`v2`	`clk_main`	`slow`	`func`	—	green

B STUDY I — PER-INSTANCE LEDGER AND MINIMAL-COMPONENT ABLATION

Every cell is re-derived from the preserved per-episode records (`flow_config.submitted.json`, `result.json`, sanitized agent log) by a committed script and is typeset directly from that script's output, so the table cannot drift from the records by transcription. The script also enforces the inclusion rule mechanically—it asserts each stage's episode count against the frozen program manifest and aborts on mismatch—because an earlier aggregation of these same records keyed cells by (condition, model) in a dictionary and so silently dropped repeat measurements of a condition; that is a failure mode a stated rule alone does not prevent. Transport is SSE streaming throughout except the DeepSeek controlled-pair rows, which are frozen non-streaming and reported as such. Program-wide: 70 episodes (58 program-primary plus 12 controlled-pair), 41 correct, 24 axis-binding, 5 role-conditioned value-selection. Minimal-component ablation (exploratory): no stable sub-component isolated—C1 alone, C2-only, C4-only and the C24 bridge all failed their pre-declared thresholds.

Per-cell uncertainty and a pooled figure we do not use. Each S0/S1/S2-M cell is one instance at k=3–4, so its exact interval is wide: BundleS 29.2–100.0% against Base 0.8–90.6% at S0 and S1, and 19.4–99.4% in both arms at S2-M. Two-sided Fisher gives $p = 0.40$ at S0, $p = 0.40$ at S1, and $p = 1.00$ at S2-M. For transparency: pooling the S0 and S1 cells into a single 2×2 table gives $p = 0.06$. We do *not* report that as a result—the design declares the task instance as the unit of analysis, and pooling two instances assumes an exchangeability the design does not establish—but we state it so that a reader can see the pooled figure was computed and set aside deliberately rather than overlooked.

C STUDY II — THE THREE STA BATCHES, REPORTED SEPARATELY

Three batches were executed on Family A, and **no quantity is ever combined across them**: nothing is summed, averaged, differenced or counted into a single n. They are (i) a three-instance pilot at

Table 4: **Study I episode ledger**, one row per (stage, condition, model): typed-binding correct/k, then the two failure subtypes. *Inclusion rule:* the 58 program-primary episodes declared by the frozen ablation-phase manifests—omitting only what those manifests mark excluded, invalid or aborted (3, 0 and 1)—plus the 12 earlier controlled-pair episodes, giving 70; this descriptive ledger is deliberately broader than the 58-episode program-primary accounting of the reproducibility statement. A condition measured in two run windows occupies two rows and is never deduplicated, which is why Base(0009)/DeepSeek appears three times and C24 twice.

stage	condition (task)	model	correct/k	axis	value
controlled pair	Base(0009)	DeepSeek	0/3	3	0
	Full bundle(0010)	DeepSeek	3/3	0	0
	Base(0009)	Qwen	1/3	2	0
	Full bundle(0010)	Qwen	3/3	0	0
C6 ablation	V1 (0013)	Qwen	3/3	0	0
	V9 (0014)	Qwen	3/3	0	0
screening	BundleD (0016)	Qwen	1/3	2	0
	BundleS (0015)	Qwen	3/3	0	0
held-out	Base(0011)	Qwen	1/3	1	1
	Held(0017)	Qwen	3/3	0	0
cross-model 1	Base(0009)	DeepSeek	2/3	1	0
	BundleS (0015)	DeepSeek	1/3	2	0
cross-model 1C	Base(0009)	DeepSeek	3/4	1	0
	BundleS (0015)	DeepSeek	3/4	1	0
decomp. 1	Contract (0019)	Qwen	1/3	2	0
	Schema (0018)	Qwen	2/3	0	1
decomp. 2	C1 (0020)	Qwen	2/4	2	0
	C24 (0021)	Qwen	3/4	0	1
decomp. 3	C2 (0022)	Qwen	1/4	3	0
	C4 (0023)	Qwen	0/4	3	1
C24 bridge	C24 (0021)	Qwen	2/4	1	1
total			**41/70**	**24**	**5**

Base = ambiguous handoff; Full bundle = C1–C7 incl. C6; V1/V9 = C6 removed from the clear instance / added to the ambiguous one; BundleS = C1+C2+C4+C7 (no C6); BundleD = C3+C5 (decoy disambiguation); Held = held-out instance + BundleS; Schema = C1+C2+C4; Contract = C7 alone; C1 = axis schema; C2 = PVT definitions; C4 = glossary; C24 = C2+C4.

k=2; (ii) a prospectively frozen twelve-instance panel at k=2; and (iii) a separately preregistered batch over those same twelve instances at k=6, which is this paper's primary S2-F evidence. Batches (ii) and (iii) share an instance set but are two independent executions: their trajectories do not form a k=8 panel, their episode counts are not added, and neither is evidence about the other's execution conditions. Where they are compared it is over per-instance *class* and *sign* only, and that comparison is labelled post hoc.

Primary batch: twelve instances at six repetitions

Statistical procedure (frozen before this batch ran). The task instance (n=12) is the independent unit and the 216 trajectories are *not* treated as n=216; pilot instances are excluded from the primary contrast; the primary contrast is the signed per-instance difference $d_i = \mathrm{BundleS}_i - \mathrm{Base}_i$; the test is the exact paired sign test, with a per-instance permutation reshuffle of the condition labels as declared secondary. Of the twelve differences 5 are non-zero and 2 of those positive, giving two-sided $p = 1.00$; the permutation sensitivity gives $p = 0.72$. Repetition depth was fixed in advance at 6 and not adapted, and no analysis was selected after seeing an outcome.

Post hoc sensitivity and panel anatomy (not preregistered). Under the same percentile bootstrap the v14 analysis applies to the k=2 batch—resampling the twelve instances with replacement, 10^5 replicates, fixed seed—the estimate -2.8pp moves over -31.9 to $+26.4$. It is a panel-composition

Table 5: **Primary S2-F panel**: per-instance outcomes over the twelve prospectively frozen instances at k=6. Correct-of-k is printed rather than a rate because every cell that is neither 0/6 nor 6/6 is one of the 7 cells whose six identical repetitions disagreed with one another; "inf." marks the 5 instances able to express a Base-vs-BundleS difference at all. Generated from the frozen report by a committed script, so the table cannot drift from the records by transcription.

	correct of k=6			difference		
inst.	Base	BundleS	TC	S−B	TC−B	class
0004	6/6	0/6	0/6	−1.00	−1.00	**inf.**
0005	0/6	0/6	0/6	0.00	0.00	floor
0006	0/6	0/6	0/6	0.00	0.00	floor
0007	5/6	0/6	0/6	−0.83	−0.83	**inf.**
0008	0/6	0/6	0/6	0.00	0.00	floor
0009	0/6	0/6	0/6	0.00	0.00	floor
0010	6/6	6/6	5/6	0.00	−0.17	ceil.
0011	0/6	6/6	6/6	+1.00	+1.00	**inf.**
0012	0/6	4/6	6/6	+0.67	+1.00	**inf.**
0013	3/6	2/6	6/6	−0.17	+0.50	**inf.**
0014	0/6	0/6	1/6	0.00	+0.17	floor
0015	0/6	0/6	2/6	0.00	+0.33	floor
mean rate	**0.278**	**0.250**	**0.361**	**−0.028**		

sensitivity band and not a confidence interval, for the reason stated below, and it is used for no hypothesis test and no decision. The anatomy is 6 floor-limited instances, 1 at the ceiling and 5 informative; dropping the two most treatment-favourable instances (n=10) moves the descriptive figure to -20.0pp. The band is *wider* than the k=2 batch's (-12.5 to $+41.7$), which is the substantive point rather than an anomaly: resolving each cell six times made the per-instance differences larger, and it is those differences that resampling perturbs. Repetition depth and panel composition are two independent limits on what such a panel resolves, and raising one does not relieve the other.

THE JOINT MODEL×FAMILY ARM: THE SAME TWELVE INSTANCES, DEEPSEEK-V4-PRO AT TWO REPETITIONS

Table 6: **S3 panel**: per-instance outcomes for DeepSeek-V4-Pro over the same twelve frozen instances at k=2 (72 episodes). At this depth a cell can only read 0/2, 1/2 or 2/2, so a 1/2 is the sole expressible form of within-cell disagreement—which is precisely why no magnitude claim is made from these values, and why this table is not to be read against Table 5 as though the two were equally resolved. 7 instances are informative; 4 sit at the floor in both compared conditions and 1 at the ceiling. Generated by the same committed script as the panel above, from the same functions.

	correct of k=2			difference		
inst.	Base	BundleS	TC	S−B	TC−B	class
0004	1/2	0/2	1/2	−0.50	0.00	**inf.**
0005	0/2	0/2	0/2	0.00	0.00	floor
0006	0/2	0/2	0/2	0.00	0.00	floor
0007	2/2	0/2	0/2	−1.00	−1.00	**inf.**
0008	0/2	0/2	0/2	0.00	0.00	floor
0009	0/2	0/2	0/2	0.00	0.00	floor
0010	2/2	2/2	2/2	0.00	0.00	ceil.
0011	0/2	2/2	2/2	+1.00	+1.00	**inf.**
0012	0/2	1/2	2/2	+0.50	+1.00	**inf.**
0013	1/2	2/2	2/2	+0.50	+0.50	**inf.**
0014	0/2	1/2	0/2	+0.50	0.00	**inf.**
0015	0/2	1/2	1/2	+0.50	+0.50	**inf.**
mean rate	**0.250**	**0.375**	**0.417**	**+0.125**		

Analysis plan (fixed and committed before any of these outcomes was read). This arm's *execution* is not covered by the k=6 study's preregistration—it ran after that study's cost gate refused a second arm, and we make no preregistration claim for it (Appendix D). What was fixed in advance is the analysis: all 72 episodes with no instance and no condition dropped; the instance as the unit (n=12), never the trajectories; BundleS $-$ Base as the primary contrast, with TypedContract secondary and explicitly non-promotable whatever the outcomes looked like; the exact paired sign test and the instance-resampling band inherited unchanged from arm 1; and the mapping from outcome to wording, so that the verdict below is a computed consequence of pre-committed thresholds rather than a phrase chosen afterwards. Support required all three of $p < 0.05$, a band excluding zero, and improvements outnumbering declines. The third holds and the first two do not: $+12.5$pp, k^+=5 of 7, p=0.45, band -16.7 to $+41.7$, permutation p=0.34 (descriptive). The licensed wording is therefore *joint transfer not established*, and the plan forbids restating it as evidence of no effect.

Post hoc: cross-model concordance in instance responses. Of the 12 instances, 10 fall in the same class under arm 1 and this arm—but 5 of those agreements are *degenerate* for the question at issue: at the floor or the ceiling in both compared conditions, they have no expressible difference in either arm, so their agreement records shared instance difficulty rather than a shared response. Only the 5 instances informative in both could have disagreed about direction, and 4 of them share the sign—the two large negative responses (0004, 0007) and both positive ones (0011, 0012) recur across the panels, while 0013, the smallest arm-1 difference, changes sign. We report this descriptively and do not launder it. The arms differ in the model *and* in k, so agreement cannot be attributed to the model alone; the jointly informative subset was identified after these outcomes were read; nothing is pooled to state it; and 4 of 5 or better would arise with probability 0.1875 if each instance's sign were an independent coin flip—a descriptive sensitivity figure, not a preregistered test, and no verdict is derived from it. The pattern is consistent with recurring task-specific structure. It does *not* establish that the instance structure is model-invariant, and it is not evidence that the aggregate effect transfers, which neither arm establishes.

EARLIER BATCH: THE SAME TWELVE INSTANCES AT TWO REPETITIONS

Table 7: Earlier STA batch: per-instance outcomes at k=2 (n=12; Y=correct, n=wrong). These are the same twelve instances as Table 5 measured in an independent earlier execution; the two tables are read side by side and never combined.

	2-rep (rate)			diff		
inst.	Base	BundleS	TC	S−B	TC−B	TC−S
0004	nY(.5)	nn(0)	nn(0)	−.5	−.5	.0
0005	nn(0)	nn(0)	nn(0)	.0	.0	.0
0006	nn(0)	nn(0)	nn(0)	.0	.0	.0
0007	Yn(.5)	nn(0)	nn(0)	−.5	−.5	.0
0008	nn(0)	nn(0)	nn(0)	.0	.0	.0
0009	nn(0)	nn(0)	nn(0)	.0	.0	.0
0010	nY(.5)	YY(1)	YY(1)	.5	.5	.0
0011	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0012	nn(0)	YY(1)	YY(1)	1.0	1.0	.0
0013	YY(1)	YY(1)	YY(1)	.0	.0	.0
0014	nn(0)	nn(0)	Yn(.5)	.0	.5	.5
0015	nn(0)	nn(0)	YY(1)	.0	1.0	1.0

Panel anatomy (post hoc; added after the frozen analysis). Table 7 also shows that the twelve instances are not twelve units of discriminating power. 6 are floor-limited—zero correct in both arms—and 1 is ceiling-limited at one in both, so 5 instances carry the entire contrast, and the two largest contributors (both moving 0/2 to 2/2) account for more than the whole aggregate ordering: dropping them leaves -5.0pp on the remaining ten. We report this because "n=12" otherwise reads as more resolving power than the panel has, and because it explains *why* the instance-resampling band of the main text is as wide as -12.5 to $+41.7$: resampling a contrast borne by five instances cannot be stable. Two cautions. This is a leave-two-out sensitivity description, not a treatment-harm finding—the two dropped instances moved cleanly and in the treatment's favour, and discarding the largest contributors from any small panel moves an estimate. And it is post hoc: it changes neither the preregistered statistic nor the verdict, which was already *not established*. Its effect is to make the aggregate ordering legible as unstable rather than as weak evidence of transfer. A prospective panel intending to resolve this contrast would need to screen instances for discriminating power before freezing, which ours did not.

Statistical procedure (frozen; descriptive sensitivity only). The task instance (n=12) is the independent unit; two repetitions are nested observations (72 trajectories are *not* treated as n=72). Pilot instances 0001–0003 are excluded from the primary contrast. Each per-instance, per-condition rate is the mean over its two repetitions. The primary contrast is the signed per-instance difference $d_i = \mathrm{BundleS}_i - \mathrm{Base}_i$. *Exact paired sign test.* Of the 12 instance differences, 5 are non-zero (3 positive, 2 negative; 7 ties), so the test is conditioned on n=5 non-zero differences with k=3 positive; the two-sided exact p-value is $2\sum_{i=0}^{\min(k,n-k)}\binom{n}{i}/2^n = 1.0$, excluding ties (standard convention). This is a non-establishment result, not evidence of a zero effect. *Per-instance permutation sensitivity.* Under the sharp null of no condition effect, we reshuffle the six within-instance condition labels (2 Base, 2 BundleS, 2 TypedContract) independently per instance, recompute $\sum_i(\mathrm{BundleS}_i - \mathrm{Base}_i)$, and report the two-sided proportion of 10^4 Monte-Carlo replicates (fixed seed 20260811) whose absolute re-randomized sum meets or exceeds the observed $\sum_i d_i = 1.5$, giving $p = 0.31$. This is a descriptive sensitivity check over the fixed instance set, *not* a population-rate p-value (the instances are not sampled from a defined population). No adaptive k was used; no trajectory-pooled p-value is reported; results are reported at the preregistered analysis.

Post hoc sensitivity (added after the frozen analysis; not preregistered). The frozen procedure above yields a test but no effect size with any indication of stability, and a p-value of 1.0 invites exactly the reading the data do not support—that the effect is zero. We therefore add a percentile bootstrap that resamples the 12 instances *with replacement* (the declared independent unit; 10^5 replicates, fixed seed), giving $+12.5$pp, and we report the *central 95% of that instance-resampling distribution*, which runs from -12.5 to $+41.7$; we call it the 95% sensitivity band. **We do not call it a confidence interval.** The declared estimand is the realized twelve-instance panel, and a realized panel's composition has no sampling distribution; resampling its members therefore does not quantify uncertainty about that estimand, it perturbs the panel. Because the draws are with replacement, each replicate duplicates some instances and omits others, so what is perturbed is the panel's empirical weighting: the band is a panel-composition sensitivity, and the 95% names a quantile range of that perturbation distribution, not a coverage probability. It is not used for hypothesis testing, is not used to change the preregistered decision rule, and licenses no claim beyond the panel.

What the band does not cover. It quantifies sensitivity to panel composition only. It does not quantify trajectory-level Monte-Carlo uncertainty: with two repetitions per instance–condition cell, each cell's latent success probability is itself poorly resolved, and that component of variability appears nowhere in the band. We deliberately do not add a third interval or a hierarchical model to absorb it—that would blur the preregistered/post-hoc boundary this appendix exists to keep sharp—and state it instead as a limit on what the band may be read to mean. It is the limit the k=6 batch was run to address, and the result of addressing it is reported above as its own batch, not folded back into this one. Every number is generated from the frozen records by a committed script and re-checked against the frozen report's own declared values.

Cross-batch structural comparison (post hoc; not preregistered; not pooling)

10 of 12 instances fall in the same class under both batches; the 6 floor-limited instances are the same 6 in each; and of the 4 instances informative in both, 4 agree in the sign of their difference. The two that change class move across a boundary by a small margin in each case (+0.5 to 0.0, and 0.0 to -0.17), which is the behavior a boundary produces rather than a contradiction. What this supports is narrow and worth stating exactly: the instance-level heterogeneity of Table 5 is a property of the *instances*, not sampling noise at k=2, because it survives tripling the repetition depth in an independent execution. What it does not support is any statement about either batch's serving stack, any combined estimate, or any claim that the aggregate direction replicated—it did not, and that contrast between a stable structure and an unstable aggregate is the observation.

Table 8: Per-instance class and signed difference under each of the two twelve-instance batches. **This is not pooling and not a backend comparison:** no quantity is summed, averaged or differenced across the two columns, and neither batch licenses an inference about the other's execution conditions. Nor is it a replication under §3.1—both batches re-run the *same* frozen instances, which is repeated measurement of stability rather than an enlargement of any projection. Both batches are classified by one function, asserted against the earlier batch's own frozen anatomy block before the comparison is formed.

	earlier batch (k=2)		this batch (k=6)		
inst.	class	S−B	class	S−B	same class
0004	**inf.**	−0.50	**inf.**	−1.00	✓
0005	floor	0.00	floor	0.00	✓
0006	floor	0.00	floor	0.00	✓
0007	**inf.**	−0.50	**inf.**	−0.83	✓
0008	floor	0.00	floor	0.00	✓
0009	floor	0.00	floor	0.00	✓
0010	**inf.**	+0.50	ceil.	0.00	✗
0011	**inf.**	+1.00	**inf.**	+1.00	✓
0012	**inf.**	+1.00	**inf.**	+0.67	✓
0013	ceil.	0.00	**inf.**	−0.17	✗
0014	floor	0.00	floor	0.00	✓
0015	floor	0.00	floor	0.00	✓

The three-instance pilot

Table 9: Three-instance STA pilot: per-instance correct-rates over two repetitions, published so the direction reversal against the twelve-instance k=2 batch (Table 7) can be checked. Generated from the frozen collection report by the same committed script that produces the interval estimates.

instance	Base	BundleS	TypedContract	BundleS−Base
0001	0.5	0.0	0.0	−0.5
0002	1.0	1.0	1.0	0.0
0003	0.0	0.0	0.0	0.0
mean	**0.500**	**0.333**	**0.333**	**−0.167**

The historical pilot (n=3) is *not* pooled with either twelve-instance batch at any point. Because the paper rests a methodological claim on the prospective expansion having reversed the pilot's descriptive direction, we publish the pilot's per-instance values (Table 9) so that the reversal is auditable rather than asserted: the pilot means are Base 0.500 against BundleS 0.333—i.e. −16.7pp, BundleS *worse*—against the twelve-instance k=2 batch's +12.5pp with BundleS better. The sign of the descriptive contrast flips. The ordering of the third condition flips too: the pilot places TypedContract level with BundleS and below Base, whereas the twelve-instance batch places it highest. Both used identical conditions, grader and two repetitions per instance–condition cell, so the reversal is not attributable to a protocol change. Read together with the k=6 batch, the pattern across all three is that the aggregate ordering has moved every time the panel or the repetition depth changed, while the per-instance structure has not.

D What would change each conclusion

§7 states that we declare in advance what evidence would move each verdict; none of it requires trusting our interpretation. These are those conditions.

- *S0–S1 (observed and replicated here) would be weakened or revised* by a preregistered counterbalanced replication on the same frozen instances that fails to reproduce the Base–BundleS separation, or by a demonstration that the bundle leaks the answer—that a solver could recover the unique assignment from BundleS without consulting the evidence. C6 is separable and the frozen condition files are released, so this is checkable—and Appendix F now reports it: a disclosure-only enumeration leaves 9 to 147 of 294 candidates under BundleS and never one, closing direct answer disclosure while leaving softer informational advantage open. What would still weaken S0–S1 is a solver recovering the assignment from BundleS *plus* generator structure, which the probe does not test.

- *S2-M (not established) would be resolved either way* by raising k on DeepSeek-V4-Pro under the same counterbalancing. Our 3/4 = 3/4 cannot distinguish an absent effect from one too small for k=4; we claim only that it is unestablished.
- *S2-F (not established on STA) could be resolved toward transfer* by a prospectively frozen panel powered to distinguish a preregistered, practically meaningful effect from zero. Our second batch raised repetition depth to 6 and did not resolve it, for a reason that constrains any third attempt: the limit was never only Monte-Carlo noise within cells but the panel's composition, so such a panel must screen instances for discriminating power *before* freezing—6 of ours sit at the floor in both arms—and preregister its target on its own grounds, since we do not treat S0–S1 as implying a cross-family effect size.
- *S3 (not established) would be resolved toward transfer* by the same remedy S2-F needs, applied jointly: a panel screened for discriminating power before freezing, and run deep enough to resolve its cells. Ours was neither. 4 of the twelve instances sit at the floor in both compared conditions, so the effective panel is 7 instances, and at k=2 each cell is estimated from two trajectories on a family where 7 of 36 cells were measured to disagree across six. A result that would revise ours is a preregistered joint panel meeting both conditions; a second run at k=2 on these same instances would not, whichever way it came out.
  How this arm came to exist, stated precisely. It is not covered by the k=6 study's preregistration, and we do not claim it is. That preregistration sized a second arm through a cost ladder, and its cost gate—evaluated on a rate pooled from a six-episode probe and one executed block—refused the arm before any of its outcomes existed. The remaining blocks were executed afterwards, at the operator's direction, which is why this arm is reported under a separate analysis plan rather than under that preregistration. What the plan fixes, and what was committed before any of these outcomes was read, is the analysis: all 72 episodes with no instance or condition dropped, the instance as the unit, BundleS versus Base as the primary contrast with TypedContract secondary and non-promotable, the sign test and resampling band inherited unchanged from arm 1, and the mapping from outcome to wording—including that a non-significant result may not be reported as evidence of no effect. Two things follow that a reader should be able to check rather than take on trust. No control was relaxed to permit the analysis: the report withholds condition aggregates if and only if a planned instance is missing—because a subset of blocks is a subset of *instances*, which would let a budget or a provider choose the sample—and this arm ran twelve of twelve, so the unmodified rule permits them. And the ordering is verifiable from the artifact's own history rather than asserted here: the plan's commit precedes the commit that brought the analysed episodes into the tree, the report records the plan's hash, and the report refuses to build at all if the plan is absent.
- *The ceiling result (SPICE) would become informative* under a harder generator for the same family: a ceiling is a property of our instances, not of the family.
- *The mechanism claim (none isolated) would be revised* by any single component reproducing the axis-suppression pattern across two independently collected run windows—a repeated-measurement test of stability, not a scope replication (§3.1). The two C24 rows of Table 4 are exactly this test, and they disagree.
- *The audit protocol's coverage claim would be strengthened or refuted* by applying the frozen rubric to a second external repair corpus; our single probe found the sampling layer entirely unsupported.

E Claim-qualification standard — full tier definitions

§3.1 states the four tiers compactly; their full definitions, including the distinctions that decide borderline cases, are these.

- **Observed.** A descriptive separation in the measured sample at that support. Carries no uncertainty claim and no target beyond E.
- **Replicated.** The direction reappears, under the same protocol, at *newly frozen configurations that enlarge the projection being claimed*: a new instance replicates in s_I, a second backend in s_M,

Table 10: Claim qualification: what a given evidence shape does and does not license. The right column collects wordings that recur in harness-effect reporting but that the evidence does not support.

Evidence shape	Licensed	Not licensed
One instance, repeated trajectories	effect observed on this instance	improves the task family
One pre-frozen held-out instance	direction replicated once	within-family generalization established
Frozen multi-instance panel	realized-panel average with a composition-sensitivity band	population transfer
Same configurations, new run window	stability of the measurement	replication at any wider scope
One alternate model backend	transfers, or fails to, on that backend	model-general effect
A family at ceiling	uninformative for this contrast	no transfer
Retrospective external coding	taxonomy relevance	taxonomy validity

an independent family in s_F. Re-running the *same* configurations in a later window is *repeated measurement*—evidence about stability, not about scope—and we never count it as replication. Because instances are family-specific, a probe that enlarges s_F necessarily enlarges s_I too, so a cross-family replication is never a replication in s_F alone.

- **Estimated on a fixed panel.** A point estimate over a declared, prospectively frozen panel, reported with an explicit account of how far it moves under resampling of that panel's members. Inference is to the realized panel and no further ($T = E$); because a realized panel's composition has no sampling distribution, what accompanies the estimate is a *sensitivity band*, not a confidence statement about a population.
- **Generalized.** Inference to a defined population of instances, models or families—the one tier asserting $T \supsetneq E$—which requires a sampling or generator distribution the design actually instantiates.

F Disclosure-only answer identifiability of the treatment conditions

This appendix supports the claim in §4 that the S0/S1 result is not direct answer disclosure; Appendix D states in advance what would move each verdict, including this one. Like Appendix G it is retrospective and post-freeze: no episode, model call or tool run, a committed script with a `--check` mode enumerating over the frozen condition files.

What is asked. Reading *only* what a condition discloses, and never the task evidence, how many of the declared candidate assignments remain compatible? Exhaustive enumeration over the declared domains gives $|\Omega| = 294$, the same universe as the worked instance of §3.1, verified byte-identical across all fourteen conditions so the counts are comparable. The probe reads `prompt.md`, `spec.md`, `glossary.md` and `public_check_summary.json`; the four evidence sources, the previous sign-off log, the shipped `flow_config.json`, the handoff manifest, the netlists and everything under `hidden/` raise an error rather than being read, and the hidden golden tuple is loaded through a separate accessor only after every survivor set is already fixed. Nine tests assert that these refusals fire, because a probe that could reach the evidence would be answering a different question.

Two readings, reported as a bracket. Whether a condition *discloses* the typed-axis membership is a judgement, so we bracket rather than choose. Under the *strict* reading a constraint counts only when its value content is stated—and BundleS's own `spec.md` says "No complete value-to-axis table is provided." Under the *leakage-favourable* reading we additionally grant the full membership table to any condition disclosing canonical labels, axis disjointness and the PVT-descriptor rule, plus clock identity from the declared v2 interface: strictly more than the treatment's files provide.

Result. BundleS leaves between 9 and 147 candidates and never one, at S0 and at the pre-frozen held-out instance alike, so the treatment cannot hand over the assignment and the observed effect

Table 11: Candidates surviving a condition's own disclosure, out of $|\Omega| = 294$. K5—the sign-off pair—is what clarity component C6 publishes, which is why the C6 rows are the probe's positive control: a component known to be answer-bearing must be detected. Neither BundleS arm is unique under either reading.

condition	role	strict	leakage-fav.	golden unique?
Base (0009)	S0 baseline	147	49	no
BundleS (0015)	S0 treatment	147	9	no
Base (0011)	S1 held-out baseline	147	49	no
Held (0017)	S1 held-out treatment	147	9	no
Base + C6 (0014)	positive control	3	1	reading-dependent
Full bundle (0010)	positive control	1	1	yes

is not literal answer disclosure. The positive control confirms the probe would have detected it: component C6 is found in exactly the two instances the ablation design declares carry it and in none of the other twelve, and it collapses the (scenario, corner) projection to one under both readings. Full-assignment uniqueness under C6 is reading-dependent—Base+C6 leaves 3 under the strict reading, because there the clock is not value-stated—so the control is stated over the projection and must not be restated as "C6 alone reduces 294 to one."

What this does not establish. Only direct answer identifiability is closed. Softer informational advantage, prior narrowing, prompt-induced heuristic cues, accidental lexical correlation and model-specific exploitation all remain open, and BundleS is *designed* to reduce semantic ambiguity: on the leakage-favourable reading it narrows the typed grid from 49 candidates to 9, a substantial advantage that stops well short of publishing the answer. The comparison is therefore not equal-information, and we do not claim it is. Nor does the probe identify a mechanism—under the strict reading BundleS does not shrink the candidate set at all relative to Base, which is why the ratio must not be restated as the treatment's causal pathway. What it does support is a structural reading consistent with the ledger's failure subtypes: the treatment converts a role-binding problem, in which which axis a label denotes is itself unknown, into a typed-selection problem over a smaller set of type-valid candidates, while leaving the evidence intersection that picks one of them entirely intact.

G SEMANTIC/TOOL DISCRIMINATION — EXCLUSION ACCOUNTING AND PAIRING SENSITIVITY

This appendix supports §3.1. It is a retrospective, post-freeze derived analysis: no episode, model call or tool run, every value re-derived from records committed at or before the experiment freeze by a committed script with a `--check` mode that recomputes and diffs against its output.

stratum	n	sem. correct	sem. wrong	tool accepts	wrong *and* accepted
workflow (program-primary)	49	33	16	49	16/16
STA, prospective panel	72	24	48	72	48/48
STA, phase-5C	12	5	7	12	7/7
STA, phase-5D	18	7	11	18	11/11
SPICE, phase-5D	18	18	0	18	— (negative control)
retained total	169	87	82	169	82/82

The retained total is shown for completeness only; it is *not* a population rate and the strata are never pooled for any reported claim.

The two verdicts. The semantic verdict is the typed oracle's: for the workflow family the submitted tuple compared to the hidden golden tuple by the same frozen classifier the Study I ledger uses; for STA and SPICE the frozen `semantic_binding` field (submitted tuple = golden). The tool-success signal is each family's own pre-existing field—`tool_exit` together with `signoff` from the stage-2 evidence record (workflow), `pt_signoff_green` (STA), `simulation_success`

(SPICE). Nothing is introduced for this analysis: a stratum recording no tool-success field is excluded, never imputed. The STA grader already computed the conjunction of interest as a marker (`green_but_unattested`) without ever aggregating it.

Exclusions: 33 of 202, never imputed. 12 controlled-pair episodes survive only as per-cell counts, with no per-episode record and therefore no tool field. 12 SPICE phase-5C episodes carry no graded tool component. 9 workflow episodes fail verdict-to-artifact pairing. We report the excluded counts rather than a bare retained total so the retained set cannot be mistaken for the whole program.

Pairing, and why it is decided by hash. A tool verdict may be compared with a semantic verdict only if both describe the same final submission. For STA and SPICE this holds by construction: the hidden runner produces the tool verdict at grading time from the submitted binding, leaving no interval in which the submission can drift. The workflow family is different—the agent runs its two-stage evidence chain during the episode and may then edit `flow_config.json`—so pairing is verified by comparing the configuration hash the stage-2 record declares it consumed against the hash of the preserved submission. Nine of 58 fail. Comparing tuples instead of hashes detects only two of the nine: the other seven agree on (scenario, corner, netlist) while the consumed file differs, and one submits a different netlist entirely. The frozen grader flags these independently through its stage-chain component, so the strict rule is primary precisely so that the derived statistic agrees with the grader rather than contradicting it.

The conclusion does not depend on the pairing rule. Under the strict hash rule the workflow stratum retains 49 episodes with 16 semantic errors; under tuple agreement, 56 with 22; with pairing not enforced, 58 with 24. In all three the tool-success signal is constant at *accept*, so every semantic error is accepted: 16/16, 22/22, 24/24. The looser rules are reported only as this sensitivity check and are not used anywhere else.

What this does and does not license. The families are deliberately constructed so a wrong binding remains tool-green, so a false-accept proportion of 1.0 demonstrates the construction operating as specified and quantifies its cost to measurement. It is not an estimate of false-accept prevalence in agent benchmarks generally, and the strata—spanning stages, conditions, models and run windows—are not one sampling frame and are never pooled into a single rate. Because the tool signal is constant, $\Delta \equiv 1 - S_{\text{semantic}}$ and is not an independent quantity. Finally, that the oracle rejects every wrong binding whenever the constraints admit exactly one satisfying assignment is a construction property of the generator, not a theorem.

H STUDY III — FULL 26-TASK TERMINAL-BENCH REPAIR CODING

Rubric frozen before coding; source: PR #53 "Terminal-Bench 2.1" (body sha256 `c7172fc6`; files sha256 `fccf2d61`); 2.0/2.1 snapshots frozen (89 tasks each; sha256 `5c7ed05c`).

Task	threat	mechanism	coverage
adaptive-rejection-sampler	Cap	instruction clarification	partial
build-pmars	Cap	instruction clarification; verifier/test tightening	partial
caffe-cifar-10	Exec; Cap	timeout; verifier/test broadening; instruction; env/dep; resource	partial
compile-compcert	Exec	environment/dependency repair	partial
configure-git-webserver	Cap	oracle repair; verifier/test tightening	partial
extract-moves-from-video	Exec	external-dependency removal	partial
filter-js-from-html	Cap; Exec	instruction clarification; resource increase	partial
fix-git	Art	hidden-artifact removal; verifier/test tightening	direct
hf-model-inference	Cap	verifier/test broadening	partial

install-windows-3.11	Cap	instruction clarification; verifier/test tightening	partial
make-doom-for-mips	Exec	oracle repair; env/dep repair	partial
mteb-leaderboard	Exec; Cap	env/dep; instruction; resource	partial
mteb-retrieve	Exec; Cap	env/dep; instruction clarification	partial
polyglot-c-py	Cap	verifier/test broadening	partial
polyglot-rust-c	Cap	verifier/test broadening	partial
protein-assembly	Cap	oracle repair	partial
rstan-to-pystan	Cap	oracle repair	partial
sam-cell-seg	Cap	instruction clarification	partial
torch-tensor-parallelism	Cap; Exec	instruction clarification; resource increase	partial
torch-pipeline-parallelism	Exec	resource increase	partial
query-optimize	Cap	instruction clarification	partial
train-fasttext	Exec; Cap	verifier/test tightening; env/dep repair	partial
build-pov-ray	Other	env/dep repair	not-covered
financial-document-processor	Other	env/dep repair	not-covered
mcmc-sampling-stan	Other	env/dep repair	not-covered
overfull-hbox	Other	env/dep repair	not-covered

Aggregate: direct 1 / partial 21 / not-covered 4; Cap 17, Exec 10, Art 1, Other 4; Samp 0; 6 multi-layer.

I Operational independence, freeze points, and phase chronology

Family independence is defined by five pre-registered structural criteria (independent templates, vocabularies, truths, graders, decoys).

Which results are exploratory and which are confirmatory. An exploratory result may not be read as a confirmation of the hypothesis it generated, so we state where each freeze fell rather than leaving it to be inferred from the narrative order.

- *Exploratory (hypothesis-generating).* Discovery of the tool-green axis-binding failure; the transport repair that made the measurement valid at all; the workflow controlled pair; and the construction of the clarity bundle. BundleS was **selected** during this stage, so its development-instance cell (S0) is exploratory by construction.
- **Freeze point 1 — intervention and held-out instance.** BundleS was fixed as C1+C2+C4+C7 with C6 withheld, and the held-out instance was frozen, *before* the held-out evaluation ran. The S1 cell is therefore a within-family confirmation of a pre-committed intervention on a pre-committed instance, which is why it is the only cell we describe as replicating anything.
- *Exploratory again.* The cross-model arm and the whole component decomposition (C1; Schema vs. Contract; C2-only, C4-only, C24 and the C24 bridge) are exploratory localization; we label them so in the main text and draw no confirmatory conclusion from them.
- **Freeze point 2 — cross-family study.** For the prospective STA study the instance panel, its size (n=12), the conditions, the randomization schedule, and the analysis (exact paired sign test, with the permutation sensitivity as a declared secondary) were all committed *before any paid call of that phase*. Sample size was not adapted and no analysis was selected after seeing outcomes. The instance-resampling sensitivity band of Appendix C was added afterwards and is labelled as post hoc wherever it appears.
- *Confirmatory.* The two prospective STA panels are this study's only preregistered confirmatory tests of transfer, and neither established an effect.

- **Freeze point 3 — the higher-replication family study.** The k=6 batch was preregistered before any of its paid calls: the same twelve frozen instances, k=6, the same conditions and grader, a frozen execution order, the analysis, and a cost gate stated as a formula. Its second arm was sized by that gate and refused by it; the S3 arm reported here was executed afterwards and is therefore governed not by this freeze point but by a separate analysis plan, committed before any of its outcomes was read (Appendix D). We keep the two labelled apart deliberately: a plan fixed before outcomes are seen is worth what it is worth, and calling it a preregistration of the experiment would claim more. Six numbered amendments exist; each was fixed before the episodes it governs entered any analysis, and none altered an analysis after an outcome was known. The cross-batch structural comparison of Appendix C was added afterwards and is labelled post hoc wherever it appears.

TypedContract entered as a *pre-specified secondary* contrast in the cross-family collection; no primary hypothesis about it was preregistered, which is why its descriptively highest rate is reported as hypothesis-generating and nothing further. The three-instance pilot preceded the twelve-instance batches and is never pooled with either.

Program accounting. Two programs, stated separately because their episodes are separate measurements. The first: episodes 58/24/36/72; costs ¥682.25 / ¥7.72 / ¥11.75 / ¥43.56, total ¥745.29. The second, the k=6 study: 216 episodes at ¥122.82 plus the S3 arm's 72 at ¥58.11, together with the discarded block passes its preregistration requires be re-run whole rather than resumed, for ¥183.93 against a ¥200 cap. That is one figure covering every ¥ the study paid, recomputed from the runner's own spend function rather than quoted from any decision record: every episode counted here stands behind a number we report. Money adds across programs; measurements do not, and no n in this paper spans both. The 58 is the workflow program-primary accounting declared by the frozen ablation-phase manifests; the 70-episode descriptive Study I ledger of Table 4 is broader because it additionally includes the 12 earlier controlled-pair episodes, which predate that accounting and its cost ledger.

J INFRASTRUCTURE + CUSTODY

Exact-commit worktree; canonical-hash guard (`FAILED_INTEGRITY` stop); episode arbiter; SSE streaming; tool-health sentinel; immutable-core/derived-deck repair; per-episode custody byte-matching. Pre-flight caught a grading-path bug (malformed signoff JSON) before the prospective run; historical pilot unaffected (0/30 triggers).

The reference-standard incident in full. §6 states the principle; this is the instance it was abstracted from. Our tool-health sentinel compares a reference artifact against a frozen fingerprint and reports the environment unhealthy on mismatch. During development it fired continuously for roughly a day, and the fault was attributed to the remote tool server. The monitor's logic was correct and its mismatch real; the attribution was wrong. The artifact it measured against had itself been rewritten—not by the remote server, but by a component of our own test harness writing into the canonical tree. The affected artifact was in the development workspace and already treated as non-authoritative, so no reported measurement derives from it; what the episode establishes is the requirement, not a result. A tripwire test now fails immediately if anything writes into the canonical dataset.

Three accounting requirements the higher-replication batch added. Each was a defect in our own bookkeeping rather than in a measurement, and each generalizes past its instance, which is why they are stated as requirements and not as an incident narrative. (1) *An unexecuted slot is not an invalid measurement.* A per-arm report initially filed slots that never ran under the same heading as episodes that ran and were discarded as measurement-invalid, inflating a collected-episode count by the size of the unrun remainder; absence of evidence must be a distinct category from evidence that was collected and rejected. (2) *A verifier that narrows its scope can pass by ceasing to look.* A cost reconciliation checked only the live execution pass, so after a legitimate whole-pass re-execution it reported agreement while no longer inspecting the pass whose understatement it had been written to catch; a check must enumerate what it claims to cover and report what it could not evaluate as a third outcome, neither pass nor fail. (3) *Raw evidence must not live on a path a later legitimate operation* *may overwrite.* Driver logs for archived passes were keyed by a reusable directory name, so a correct re-run destroyed the source a recorded figure had been derived from—the figure remained checkable only because it had been transcribed into a committed record first. None of these changed a reported number; each would have permitted one to go unchecked.

K Positioning relative to prior work

Table 13: Positioning relative to related work. The right column states what this paper adds beyond each line of prior work; the Related Work section gives the same positioning in prose for the works it cites, and the paragraph below covers the three closest concurrent works and the transfer-dimension literature.

Prior work	What it establishes	What this paper additionally does
Harness-Bench (Yao et al., 2026)	Broad model×harness characterization	Claim-scope lattice + controlled semantic mechanism
Rethinking (Wang et al., 2026a)	Leaderboard conflation; search/overfitting	Scope coordinates are partially ordered, not nested
HarnessOpt-Bench (Ursekar et al., 2026)	Optimizer capability	Studies validity of harness-effect claims
ABC (Zhu et al., 2025) / Protocol validity (Shao et al., 2026)	Setup; shortcut/reward validity	Per-control threat log
AFTER (Belikova et al., 2026)	Local/cross-task/cross-role/cross-model transfer of procedural skills	Claim indexed by its configuration support; unmeasured separated from unestablished

The three closest concurrent works, and the transfer-dimension literature. Liu et al. (2026) audits trajectory-level safety properties of harness execution (boundary compliance, execution fidelity, system stability; 210 tasks across 8 domains); our object differs—we audit the evidentiary scope and measurement validity of semantic-binding performance claims, not execution-trajectory safety. Raj et al. (2026) localize a failure to a model/harness/tool/environment interaction edge—*where* a failure originates, where our layers ask whether the *measurement* of it was valid. Caban (2026) argue from psychometric validity that agentic evaluation compounds reliability failures, the concern our sampling and execution layers instantiate empirically. None indexes the evidence for a harness-effect claim by the support it occupies. Reporting transfer along several named dimensions is not itself new either: Belikova et al. (2026) evaluate procedural-memory skills under separate local, cross-task, cross-role and cross-model settings, finding that some transfer broadly while others specialize to a role. Our contribution is not that observation but the indexing—locating the *evidence* for a claim by the set of configurations it occupies, which separates a hypothesis measured and left unestablished from one never measured at all, together with a qualification standard tied to that index.

Ethics statement. The preregistered blinded human construct-validity study (Study B) is preregistered but unexecuted because qualified independent human annotators were unavailable; no LLM annotator was substituted.

AI-use statement (outside the page limit). Generative-AI assistants were used to support experimental design and critique, orchestration and infrastructure code, task generators, analysis and report scripts, literature search, result-interpretation checks, and manuscript drafting and editing. All experimental decisions, task definitions, acceptance criteria, statistical analyses, and scientific claims were reviewed and approved by the authors. Experimental results were produced by frozen executable pipelines and independently checkable graders reproduced from frozen manifests; generative-AI systems were not used as ground-truth judges or scientific oracles. The authors take responsibility for all content and conclusions.

Reproducibility statement. Deterministic generators (fixed seeds); independent per-family graders; exact-commit isolated-worktree execution with pre/post-episode canonical-hash verification; committed sample-membership arbitration; validity-only replacement; per-episode custody byte-matching.

Program totals, kept separate because their episodes are separate measurements: 58+24+36+72 paid primary episodes at a committed-ledger cost of ¥745.29, and the later k=6 study's 216 episodes plus its S3 arm's 72 at ¥183.93 against a ¥200 cap. (The descriptive Study I ledger of Table 4 reports 70 workflow episodes: the 58 plus the 12 earlier controlled-pair episodes, which sit outside that program accounting and its cost ledger.) The second study's preregistration, its six numbered amendments, and the separate pre-outcome analysis plan governing its S3 arm are released with it, together with the commit history that places the plan before the analysis, so the ordering is auditable rather than asserted. Frozen manifests, randomization schedules, custody hashes, per-episode sanitized evidence, and the scripts that re-derive every table from the preserved records are in the anonymized supplement; every table in this paper is emitted by one of those scripts, each of which has a `--check` mode that recomputes and fails on drift.